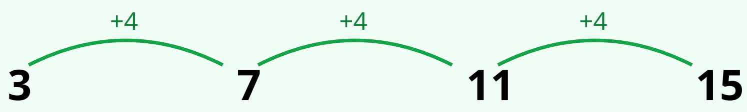
Quadratic Sequences

AQA · KS3 Year 7–8

Name: ___________________ Class: _________ Date: _________

Recognise a quadratic sequence by its constant second difference, and find the nth term in the form $an^2 + bn + c$ using the second-difference method.

VISUAL EXAMPLE



nth term = 4n - 1

Arithmetic sequence: a constant difference is added each step.

COMMON MISTAKES

- Forgetting to HALVE the second difference to find a (the coefficient of n^2).
- Making arithmetic slips when subtracting the an^2 values from the sequence.
- Not finding the leftover linear part ($bn + c$) correctly.
- Sign errors, especially when the linear part is negative.

METHOD

1. Find the first differences between consecutive terms.
2. Find the second differences — for a quadratic these are constant.
3. a = second difference ÷ 2; this gives the an^2 part.
4. Subtract an^2 from each term to leave a linear sequence; find its nth term ($bn + c$).
5. Combine to get $an^2 + bn + c$ and check by substituting n = 1, 2.

WORKED EXAMPLE

Find the nth term of: 3, 8, 15, 24, 35, ...
Step 1 — First differences: 5, 7, 9, 11.
Step 2 — Second differences: 2, 2, 2 (constant)   quadratic.
Step 3 — $a = 2 \div 2 = 1$, so the n^2 part is n^2.
Step 4 — Subtract n^2 (1, 4, 9, 16, 25): 2, 4, 6, 8, 10 = 2n.
Step 5 — nth term = $n^2 + 2n$ (check n = 1: 1 + 2 = 3   )

1 Find the first and second differences of each sequence. (2 marks)

a) 2, 5, 10, 17, 26, ...

b) 4, 7, 12, 19, 28, ...

HOW TO START

First differences are the gaps between terms; second differences are the gaps between those.

For 2, 5, 10, 17, 26: first differences 3, 5, 7, 9; second differences 2, 2, 2.

ANSWERS

(a) = __________

(b) = __________

METHOD 1 First differences | 2 Second differences (constant?) | 3 a = second difference ÷ 2 | 4 Subtract an^2 find bn + c | 5 Combine: $an^2 + bn + c$

2 State whether each sequence is quadratic. Show the second differences. (2 marks)

a) 1, 4, 9, 16, ...
b) 2, 4, 6, 8, ...

HOW TO START

A sequence is quadratic if its SECOND differences are constant.
1, 4, 9, 16 has second differences 2, 2 quadratic.

ANSWERS

(a) = ____________________

(b) = ____________________

METHOD 1 First differences | 2 Second differences (constant?) | 3 a = second difference ÷ 2 | 4 Subtract an^2 find bn + c | 5 Combine: $an^2 + bn + c$

3 Find the nth term of each quadratic sequence. (3 marks)

a) 2, 5, 10, 17, 26, ...
b) 4, 7, 12, 19, 28, ...
c) 3, 9, 19, 33, 51, ...

ANSWERS

(a) = ____________________

(b) = ____________________

(c) = ____________________

METHOD 1 First differences | 2 Second differences (constant?) | 3 a = second difference ÷ 2 | 4 Subtract an^2 find $bn + c$ | 5 Combine: $an^2 + bn + c$

4 A sequence has nth term $n^2 + n$. (3 marks)

a) Find the 1st term.
b) Find the 5th term.
c) Find the 10th term.

ANSWERS

(a) = ______

(b) = ______

(c) = ______

METHOD 1 First differences | 2 Second differences (constant?) | 3 a = second difference ÷ 2 | 4 Subtract an^2 find $bn + c$ | 5 Combine: $an^2 + bn + c$

5 Find the nth term of each quadratic sequence. (3 marks)

a) 0, 3, 8, 15, 24, ...
b) 5, 14, 27, 44, 65, ...
c) 3, 8, 15, 24, ...

ANSWERS

(a) = ______________________

(b) = ______________________

(c) = ______________________

METHOD 1 First differences | 2 Second differences (constant?) | 3 a = second difference ÷ 2 | 4 Subtract an^2 find $bn + c$ | 5 Combine: $an^2 + bn + c$

6 A growing pattern of squares uses 2, 6, 12, 20, ... tiles for patterns 1, 2, 3, 4. (3 marks)

a) How many tiles are in pattern 5? (1 mark)

b) Find the nth term for the number of tiles. (2 marks)

ANSWERS

(a) = ______________________

(b) = ______________________

METHOD 1 First differences | 2 Second differences (constant?) | 3 a = second difference ÷ 2 | 4 Subtract an^2 find $bn + c$ | 5 Combine: $an^2 + bn + c$

7 A quadratic sequence is 1, 6, 15, 28, 45, ... (4 marks)

a) Show that the second difference is constant. (1 mark)

b) Find the nth term. (2 marks)

c) Find the 10th term. (1 mark)

ANSWERS

(a) = ______

(b) = ______

(c) = ______

METHOD 1 First differences | 2 Second differences (constant?) | 3 a = second difference ÷ 2 | 4 Subtract an^2 find bn + c | 5 Combine: $an^2 + bn + c$

8 A quadratic sequence has nth term $n^2 - 3n + 2$. (5 marks)

a) Find the first four terms. (2 marks)

b) Show that the second difference is 2. (1 mark)

c) Which term has the value 30? (2 marks)

ANSWERS

(a) = ______

(b) = ______

(c) = ______

METHOD 1 First differences | 2 Second differences (constant?) | 3 a = second difference ÷ 2 | 4 Subtract an^2 find $bn + c$ | 5 Combine: $an^2 + bn + c$

Self-reflection

How confident do you feel with this topic now? (tick one)

Not yet | Getting there | Confident | Really confident

Which question did you find the hardest, and why?

Why do I halve the second difference?

One thing I will practise before the next lesson:

Teacher key

Q1: a) first 3, 5, 7, 9; second 2, 2, 2 b) first 3, 5, 7, 9; second 2, 2, 2

Q2: a) second differences 2, 2 quadratic b) first differences all 2, second differences 0 not quadratic (it is linear)

Q3: a) a = 1 n^2; subtract gives 1, 1, 1… $n^2 + 1$ b) $n^2 + 3$ c) second difference 4 a = 2 ($2n^2$); subtract gives 1,1,1… $2n^2 + 1$

Q4: a) $1^2 + 1 = 2$ b) 25 + 5 = 30 c) 100 + 10 = 110

Q5: a) $n^2 - 1$ b) second difference 4 a = 2; subtract $2n^2$ gives 3, 6, 9… = 3n $2n^2 + 3n$ c) $n^2 + 2n$

Q6: a) 30 tiles b) second difference 2 a = 1 (n^2); subtract gives 1, 2, 3, 4 = n $n^2 + n$

Q7: a) first differences 5, 9, 13, 17; second differences 4, 4, 4 (constant) b) a = 2 ($2n^2$); subtract gives -1, -2, -3… = -n $2n^2 - n$ c) 2(100) - 10 = 190

Q8: a) 0, 0, 2, 6 b) first differences 0, 2, 4; second differences 2, 2 c) $n^2 - 3n + 2 = 30$ $n^2 - 3n - 28 = 0$ $(n - 7)(n + 4) = 0$ n = 7 (the 7th term)